INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

SPRAYING UNIT, CLEANING-DEICING- ANTI-ICING FLUID, TRUCK MOUNTED

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all Federal agencies.

1. SCOPE. This CID describes chassis mounted, diesel powered, spraying units, which are capable of deicing, anti-icing, cleaning, and decontaminating the entire range of airframes in the U. S. Air Force's inventory. Two general classifications are herein defined: (1) a basic unit, Class A, that can de/anti-ice and clean all but the upper tail sections of C-5 and C-17 airframes; and, (2) a spraying unit, Class B, provided with the extended reach necessary to effectively de/anti-ice C-5 and C-17 horizontal stabilizers. Both Classes of spraying units include boom-mounted enclosed cabs from which de/anti-icing and decontamination operations are performed. The spraying units are capable of the controlled application of heated deicing fluid, deicing fluid delivered via forced air and/or high efficiency nozzles, and unheated anti-icing fluid. An auxiliary power transmission source such as an engine or hydrostatic drive is provided to hydraulically operate all de-icing components including boom and enclosed cab actuation, deicing-fluid heater motor and fuel pump, and separate deicing and anti-icing pump motors.

2. CLASSIFICATION. The two Classes of spraying units covered by this CID are classified as follows:

Class	Elevation of boom	Minimum Tank Capacity
A	42 feet	1,600 gal
B	75 feet	2,600 gal

3. SALIENT CHARACTERISTICS.

3.1 Spraying unit description. Both Classes A and Class B spraying units shall be constructed in accordance with the requirements of SAE ARP-1247, General Requirements for Aerospace Ground Support Equipment, Motorized and Non-motorized, SAE ARP-1971, Aircraft Deicing Vehicle – Self-Propelled, Large Capacity, and this document. In the event of a conflict between either ARP-1247 or ARP-1971 and the text of this document, this document shall take precedence. Nothing in this document, however, overrides applicable laws and regulations, unless specific

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FSC 1730

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exception has been obtained by the procuring agency. Unless otherwise specified, each of the following salient characteristics shall be applicable to both Classes A and Class B spraying units. The spraying unit shall be designed for two-person operation or, when specified, one-person operation. It shall be maintainable with standard tools and built with modular parts supplied from commercial sources.

3.1.1 Modified commercial product. The spraying unit, at a minimum, shall be the manufacturer's standard commercial product modified as necessary to be in accordance with the requirements specified herein. Additional or complimentary features which are not specifically prohibited by these specifications, but which are part of the manufacturer's standard product, shall be included in the unit being furnished.

3.1.2 Components: The spraying unit shall consist of the following major components:

Item or System Name	See Characteristic
Hydraulic system	3.11
Chassis	3.12
Truck cab	3.13
Unit frame	3.14
Boom assembly	3.15
Stabilizers	3.16
Enclosed cab	3.17
Auxiliary power source	3.18
Plumbing	3.19
Pumping systems	3.20
Fluid heating system	3.21
Fluid tanks	3.22
Delivery system	3.23

3.2 Design and construction. The spraying unit shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components. The controls shall be readily operable by personnel wearing heavy, arctic gloves and clothing, or mission-oriented protective posture (MOPP) level 4 chemical warfare gear.

3.2.1 Materials, protective coatings, and finish.

3.2.1.1 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the item shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Fasteners, handles, and fittings used in the assembly of the item shall also be primed and painted.

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3.2.1.2 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of dissimilar base metals to provide electromotive-compatible abutting surfaces is acceptable. The use of dissimilar metals is permitted only when separated by suitable insulating material, except in systems where bridging of insulation materials by an electrically conductive fluid can occur. Sealants or gel-type gasket materials shall be used between faying surfaces and butt joints.

3.2.1.1.1 Rustproofing. The vehicle chassis and cab (at minimum: seams, welds, locations where dissimilar metals contact, etc.) shall be rustproofed and undercoated in accordance with FED-STD-297. When specified, the level of corrosion prevention shall be upgraded to tropical (see 7.2.b). Tropical protection shall be corrosion-protected with a thick, hard, black, commercially available, corrosion-preventive compound (CPC) that is difficult to remove. Application shall include the exposed chassis-frame metal underneath the spraying unit (excluding the drive train), the inside of the wheel wells, and behind the fender skirts. In addition, spraying unit that are to be shipped overseas shall be temporarily corrosion-protected from the effect of salt spray, both on the ship and at port, with the application of a thin, soft-film, semi-transparent, commercially-available CPC that is easily removable by washing or with low-pressure steam. Application of temporary CPC shall be to the exterior of the spraying unit, in critical or corrosion-sensitive areas of the vehicle. The Air Force vehicle maintainers overseas shall remove this temporary CPC on receipt from the shipper. The technical manual shall reflect the temporary CPC removal process.

3.2.1.3 Finish. Unless otherwise specified (see 7.2.c), the exterior finish color of the spraying unit shall be Semi-gloss Green, Color Number 24052 of SAE AMS-STD-595. Exterior surfaces shall be prepared, primed, and painted with polyurethane paint. The standard paint shall be resistant to aircraft hydraulic fluid. The interior of all compartments shall be painted with an impact-resistant, textured coating that resists stains, scuffs, chips, and scratches. All exterior surfaces, excluding all normally bright metal and anodized parts, shall be painted body color. The chassis frame and running gear may be black. All interior surfaces visible with any compartment door open, excluding the interior of the cab, shall also be painted body color. This includes compartment shelves and mounting hardware, but does not include items mounted in the compartments.

3.2.1.4 Exclusion of water. The design of the spraying unit shall be such as to prevent water leaking into, or being driven into, any part of the spraying unit interior when either in an operating or travelling configuration. All windows, doors, panels, covers, and alike items, shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Particular care shall be taken to prevent wetting of equipment and heat and sound proofing materials. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage shall be provided for storage compartments, engine compartments, and other areas in the spraying unit that could collect and retain water.

3.2.1.4.1 Fluid traps and faying surfaces. There shall be no fluid traps on the spraying unit. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

3.2.1.4.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

3.2.1.4.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The number and location of drain holes

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shall be sufficient to permit drainage of all fluids when the unit is on a one degree incline in any plane. The minimum size of the drain holes shall be 0.25 inch.

3.2.2 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible, but shall not be less than ½-inch high. Markings, Information/Caution shall be Lusterless Black, Color Number 37038 of SAE AMS-STD-595, and Markings, Warning/Danger shall be Lusterless Red, Color Number 31136 of SAE AMS-STD-595. The center of gravity of the spraying unit shall be stenciled on the unit within 1.0 inch of the calculated center of gravity.

3.2.3 Identification and information plates.

3.2.3.1 Identification plate. An identification plate in accordance with MIL-STD-130 shall be securely attached to the spraying unit in a readily accessible location. The identification plate shall contain the following information: nomenclature, part number, serial number, date of manufacture, manufacturer's name, Commercial and Government Entity (CAGE) code, date of warranty expiration, and National Stock Number (NSN). The spraying unit, and any of its components for which the government's unit cost is more than $5,000, is serially managed, or the procuring agency determines is mission essential, shall have Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near the respective identification plate(s), marked in accordance with MIL-STD-130. UID information shall be included as both a bar code and human readable markings. The European Conformity, "CE" marking shall be affixed in accordance with European Union requirements on or adjacent to the identification plate.

3.2.3.2 Transportation data plate. A transportation data plate shall be securely attached to the spraying unit in a readily accessible location. The plate shall contain at least the following information:

a. Side and rear silhouette views of the spraying unit.

b. Horizontal and vertical location of the center of gravity of the spraying unit in air transportable configurations for C-17 and C-5 aircraft, marked on the silhouette views.

c. Shipping weight.

d. Loading cubage.

e. Overall height, width, and length.

f. Front and rear axle loads.

g. Tie down information.

3.2.3.3 Lubrication data plate. The spraying unit shall be provided with a permanently-marked lubrication data plate constructed of a non-corroding metal, mounted on the exterior surface of the spraying unit that shall direct attention to all lubrication fittings and components that require lubrication. The plate shall identify the type and grade of lubricant required for operational temperatures specified in 3.3.1.

3.2.4 Safety. In addition to the safety requirements specified in 3.8, 3.9, and 3.10 of SAE ARP

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1247 the following safety features shall be included:

a. Emergency shutdown control switches shall be provided in the enclosed cab, truck cab control panel, and by the ground control station to shut down the auxiliary power source fluid and heating system as well as stop all boom movement. The communication system (see 3.17.1) shall not be interrupted by the activation of this switch.

b. Automatic shutdown of pumps in the event of low fluid level.

c. A fire extinguisher system, which includes a mechanism that shuts down the fluid heater upon activation. The system being installed must be certified by the manufacturer for its intended use. The system shall be automatically activated as well as manually operated, as per 3.9.9 of SAE ARP 1971. This system shall have a minimum inspection requirement of "annual" and no special certification should be needed for the inspection of the system.

d. Shutdown devices with manual reset switches in the event of overheating or misfiring of the fluid heater.

e. Automatic 45-second fresh air purge of heater before ignition and manual 150-second fresh air post-purge after heater shut down if the unit contains an air compressor.

f. Interlock system to prevent driver from operating in excess of 5 mph while the boom is in the raised, out of boom rest, or cradle position.

g. Means of regaining control of the enclosed cab and boom from ground level in the event of disabling accident or injury to operator in the enclosed cabin by way of a 12-volt (DC) emergency hydraulic pump.

h. Redundant proximity switches shall be provided that identify maximum boom extension and when stabilizers are required in the lower envelope (Class B only).

i. Redundant proximity switches shall be provided that identify minimum boom angle in the upper envelope (Class B only).

j. Audible backup warning alarm.

k. Audible stabilizer deployment alarm (Class B only), which may be the same as the back-up alarm. This alarm should be located at ground level to be heard by personnel in close proximity.

l. A red or amber flasher, meeting minimum aviation requirements (FAA AC 150/5210-5), shall be installed on the rotator assembly at the highest and most visible point. A flasher control switch shall be located on the truck cab control panel.

m. Warning decals, as appropriate.

n. Chassis breaks shall lock out in an emergency, with or without power.

3.2.4.1 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion

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and shall be located so as to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

3.2.4.2 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the spraying unit with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

3.2.4.3 Sound levels. The spraying unit sound levels shall conform to the recommended practice indicated in SAE J 1096.

3.2.4.4 Upper deck access. Upper deck access shall be designed in accordance with applicable sections of Operational Safety and Health Administration (OSHA) 29 CFR 1910 in addition to safety requirements listed in 3.2.4 of this document.

a. Upper deck access of the vehicle for maintenance purposes shall be provided.

b. Two Fall Prevention Anchorage points with a minimum capacity rating of 5,000 pounds (lbs) each shall be provided. The inside measurement or diameter of the anchorage point shall be a minimum of 1 inch. Location of anchorage points shall allow for uninhibited safe access to upper deck components and must be approved by the Procuring Activity. Specifics found in OSHA 29 CFR 1910 .66 App C.

c. A guard for the auxiliary engine harmonic balancer and pulley assembly shall be provided to prevent personnel from coming into contact with rotating and/or reciprocating parts.

3.2.5 Human engineering. The spraying unit shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens and mission-oriented protective posture (MOPP) level 4 chemical warfare gear.

3.2.6 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial components. Tapped threads shall have a minimum thread engagement in accordance with Table I.

TABLE I. Minimum thread engagement.

Material	Minimum Thread Engagement
Steel	1.0 times the nominal fastener diameter
Cast iron, brass, or bronze	1.5 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

3.2.7 Welders and welding. All welders shall be certified to weld in accordance with AWS D1.1 and AWS D1.2, as applicable. The contractor shall make available to the government certifications for all welders being utilized on the spraying unit. Welding procedures and all welding on the spraying unit shall be in accordance with AWS D1.1 and AWS D1.2, as applicable. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall

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transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

3.3 Environmental conditions.

3.3.1 Operating temperature range. The spraying unit shall be capable of operating in ambient temperatures ranging from -40 to 125 °F.

3.3.2 Storage temperature range. The spraying unit shall be capable of being stored in ambient temperatures ranging from -40 to 125 °F.

3.3.3 Precipitation.

3.3.3.1 Rain. The spraying unit shall be capable of storage and operation during rainfall of 5 inches per hour for three consecutive hours and 10 inches per hour for 10 consecutive minutes, with winds of up to 35 knots; and with 6 inches of rain per hour impinging on the spraying unit at angles from vertical to 45°.

3.3.3.2 Snow. The spraying unit shall be capable of storage and operation during accretion of wet snow up to 2 inches per hour for at least 12 hours.

3.3.3.3 Ice. The spraying unit shall be capable of storage and operation with ice accretion up to 1.5 inches on exposed horizontal surfaces. An operator may use an ice scraper for five minutes during the start-up process.

3.3.4 Solar radiation. The spraying unit shall not be adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.

3.3.5 Fungus. All materials used in the spraying unit shall be fungus resistant or shall be suitably treated to resist fungus. Materials treated for fungus resistance shall retain their original electronic and physical properties, shall not present toxic hazards, and treatment shall last for the entire service life of the part.

3.3.6 Salt fog. The spraying unit shall be capable of storage and operation in high temperature, high humidity, salt laden, and seacoast environments without damage or deterioration of performance.

3.3.7 Sand and dust. The spraying unit shall be capable of storage and operation during exposure to wind-blown sand or dust without damage or deterioration of performance.

3.4 Weight and dimensions. Overall weight and dimensions in air transport configuration (see 3.5.2) shall not exceed:

Weight	60,000 pounds
Length	496 inches
Width	102 inches (windows retracted)
Height	153 inches

3.5 Transportability.

3.5.1 Surface transportability. The spraying unit shall be transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366, and shall be capable of

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withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport, except that design for rail impact testing (see 5.2.5 of MIL-STD-1366) is not required.

3.5.2 Air transportability. The spraying unit shall be transportable on C-5 and C-17 cargo aircraft in accordance with MIL-STD-1791. The configuration of the spraying unit for air transport may be reconfigured (parts folded in, down, or as needed) but may not be removed. In all air transport configurations, the spraying unit shall be capable of being restrained and withstanding, without loss of serviceability, 2.0 G up and 4.5 G down accelerations, and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The spraying unit shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. The spraying unit shall roll on and off the aircraft, negotiating the required maximum ramp angles without shoring.

3.5.2.1 Shoring. The spraying unit shall be air transportable without shoring.

3.5.2.2 Axle weight. Axle weight shall not exceed 20,000 pounds.

3.5.2.3 Tire pressure. Tire pressure shall not exceed 100 pounds per square inch (psi) and shall not be reduced for air transport.

3.5.2.4 Equipment removal and reconfiguration. Preparation for air transport shall take no more than 2 man-hours and restoration to operating configuration shall take no more than 2 man-hours for 2 persons using common hand tools (see 7.4.1). All equipment removed shall be stored on the spraying unit; caps and plugs shall permit moving and storage in transport configuration.

3.5.3 Tie downs. The spraying unit shall be symmetrically restrained during air and ground transport. Tie down points shall be rated at a minimum of 25,000 pounds, marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the spraying unit. The tie down provisions shall be in accordance with 4.1 through 4.12 of MIL-STD-209.

3.6 Maintainability. The spraying unit shall be designed for maintainability in accordance with 5.9 through 5.9.18 of MIL-STD-1472; forces shall not exceed those specified for both males and females. The spraying unit shall be designed and constructed as follows:

a. There shall be a minimum number of parts consistent with reliability and performance specified herein.

b. All de/anti-icing and decontamination components shall be readily accessible, either by hinged compartment doors or removable panels, which shall not require more than five (5) minutes to open or remove.

c. The spraying unit shall allow adjustments, servicing, maintenance, and replacement of parts and components with minimum disturbance to other equipment, parts, or components.

d. Parts and components shall be positioned for access and ample working space shall be provided, unless performance or reliability is appreciably degraded by the access location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be provided primary access.

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e. The number of tools required to accomplish maintenance shall be minimized by applying design practices such as reducing the variety of bolt head sizes.

f. Where possible, all wiring shall be incorporated into harnesses and replaceable within 30 minutes.

g. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.6.1 Preventive maintenance. The recommended preventive maintenance interval (PMI) shall be at least 200 operating hours, or 8 weeks, whichever occurs first. Daily preventive maintenance tasks shall not require more than 1 man-hour. Other important maintenance intervals include weekly, 250 hour/monthly, and seasonal maintenance checks. The weekly preventative maintenance tasks shall not require more than 2 man-hours and the 250 hour/monthly and seasonal preventative maintenance tasks shall not require more than 8 man-hours.

3.6.2 Inspection and servicing provisions.

a. Pre-use inspections shall not require hand tools

b. Drain plugs and filters shall be directly accessible from the ground and oriented to have unimpeded drainage to a catch pan.

c. The spraying unit shall be designed with maximum usage of sealed lifetime lubrication bearings.

3.6.3 Special tools. The design of the item shall minimize the requirement for special tools (see 7.4.2). All special tools shall be provided with, and stored on, the spraying unit.

3.6.4 Diagnostic software. A copy of any diagnostic software required or recommended for maintaining the spraying unit shall be provided with each spraying unit on CD-ROM or DVD-ROM.

3.6.5 Special test equipment and software. When specified (see 7.2.d), the spraying unit shall come shipped with any software or equipment needed to perform any testing not able to be completed with conventional software or common hand tools (see 7.4.1). All delivered software shall function as intended and shall be free of vulnerabilities (i.e. defects, bugs, malware, etc.), either intentionally or unintentionally or inserted as part of the software throughout the lifecycle. Any interface hardware/cable required to communicate between the host computer with diagnostic software and the spraying unit shall also be provided with each spraying unit.

3.7 Performance. The spraying unit shall be capable of providing full deicing and anti-icing operations under the following conditions without detrimental effect to subsequent operation:

a. Sea level.

b. Elevations up to 6,000 feet above a relative humidity of at least 100 percent at an operating temperature of 125 °F.

c. Wind loads of at least seven (7) pounds per square foot.

d. Continuous 25 miles per hour (mph) operating speed on horizontally level surfaces with fluid tanks full in accordance with 3.2.2 of SAE ARP 1971.

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e. Continuous 4 miles per hour operating speed when boom of Class A spraying unit is not cradled or when boom of Class B spraying unit is not cradled and outriggers/ stabilizers are not deployed.

3.8 Mobility. The spraying unit shall be in accordance with SAE ARP 1247, except as otherwise specified herein.

3.8.1 Terrain.

3.8.1.1 Paved surfaces. The Class A spraying unit shall be drivable over paved surfaces at speeds up to 25 miles per hour (mph) when fully loaded and speeds up to 45 mph when unloaded or empty. The Class B spraying unit shall be drivable over paved surfaces at speeds up to 25 mph when fully loaded and unloaded.

3.8.1.2 Graded gravel. The loaded and unloaded Class A and Class B spraying units shall be drivable at speeds up to 10 mph over graded gravel.

3.8.2 Towing force. The towing force shall be in accordance with 3.13.1.3.15 of SAE ARP1247.

3.8.3 Clearances. The spraying units shall have a ground clearance, ramp break-over angle, angle of approach, angle of and departure in accordance with 3.13.1.7.1.8 of SAE ARP1247.

3.8.4 Transmission. The driveline shall include an automatic transmission. The net torque capacity and the net power rating of the transmission shall exceed the output ratings of the engine.

3.8.5 Suspension. The spraying unit shall have a suspension system in accordance with 3.13.1.7.1.10 of SAE ARP1247.

3.8.6 Steering. The spraying unit shall have a steering system in accordance with 3.13.1.7.1.15 of SAE ARP1247.

3.8.7 Brakes.

3.8.7.1 Parking brakes. The spraying unit shall have parking brakes in accordance with 3.13.1.7.2.3 of SAE ARP1247.

3.8.7.2 Service brakes. The spraying unit shall have service brakes in accordance with 3.13.1.7.2.5 of SAE ARP1247.

3.8.8 Bumper. The spraying unit shall have a rear bumper in accordance with ANSI/ITSDF B56.1.

3.8.9 Fenders. The spraying unit shall have fenders in accordance with 3.13.1.7.1.16 of SAE ARP1247.

3.8.10 Mud-flaps. The spraying unit shall have mud-flaps in accordance with 3.13.1.7.1.16 of SAE ARP1247.

3.8.11 Tires and wheels. The spraying unit shall be equipped with single or dual tubeless steel belted radial all-season tires with on/off-road type tread mounted on steel disc wheel assemblies. Tire and wheel assemblies shall be identical at all wheel positions. Tires and wheels shall be in accordance with the *Tire and Rim Association Year Book* requirements for this application.

3.9 Engine and related equipment. The spraying unit shall be equipped with a diesel engine. The engine shall be certified to comply with the most stringent Environmental Protection Agency (EPA) non-road diesel engine emission requirements possible while meeting the requirement to operate on fuels containing over 15 parts per million (ppm) sulfur.

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3.9.1 Export engine. When required, the spraying unit shall be equipped with a diesel engine compatible with JP-8 and F-24 jet fuels (see 3.9.7.1). At a minimum, the engine shall meet a Tier 2-level exhaust and evaporative emissions standard (export engine equivalent) for on-highway engines. This represents the highest level of exhaust and evaporative emissions standards that will permit the use of high-sulfur diesel fuels. In accordance with Title 40 of the Code of Federal Regulations (40 CFR), 85.1708, the Air Force possesses an EPA waiver to allow for deviation from EPA emissions guidelines. The waiver will be transferred from the Air Force to the engine manufacturer. The transfer process requires the engine manufacturer to supply a decal for the engine stating, "This engine has an exemption for national security under 40 CFR, 1068.225."

3.9.2 Engine starting system.

3.9.2.1 Starter. The engine shall be equipped with a 12-volt DC electric starter.

3.9.2.2 Engine starting aids. All engine starts shall be within 15 seconds cranking in any ambient temperature; within the required operating range of the spraying unit. Internal engine starting aids, fluid starting aids, and heat from the winterization system (see 3.10.3) may be used prior to and during the start period to facilitate engine starting under the following conditions in Table II:

TABLE II. Engine starting aids.

Temperature Range	Starting Aids Permitted
40 through 130 °F	None
0 through 39 °F	Internal engine starting aids
-40 through -1 °F	Internal engine starting aids and heat from the winterization system

3.9.3 Engine air intake system. The engine air intake system shall be in accordance with 3.13.1.4.3 of SAE ARP1247. The inlet shall not draw air from directly beneath the spraying unit. The inlet shall not be located near the cooling system air outlet nor the engine exhaust outlet. Joints shall be minimized between the air filter outlet and the actual engine air inlet. The air intake system shall be designed to ensure no leakage of unfiltered air into the engine. A differential pressure air filter service indicator shall be provided.

3.9.4 Engine cooling system. The engine cooling system shall be in accordance with 3.13.1.4.2 of SAE ARP1247. Silicone radiator and heater hoses, constant-torque clamps, a coolant filter, and a coolant recovery system shall be provided. Engine coolant shall be in accordance with A-A-52624, Type I, and of adequate strength to provide protection to -40 °F. The engine out (top of radiator) coolant temperature shall not exceed 210 °F. The coolant filter shall be rated for one year or 2,000 hours between replacements. Supplementary coolant additives (SCA) shall be included in the filter or added separately.

3.9.5 Engine lubrication system. The engine lubrication system shall be designed so that the spraying unit can be operated on a one degree incline in any plane.

3.9.5.1 Engine oil. The engine shall be compatible with Grade 15W40 of MIL-PRF-2104 from 0 to 250 °F. Oil pre-heat for operation below 0 °F is allowed. The engine shall be compatible with arctic engine oil in accordance with MIL-PRF-46167 from -40 to 60 °F.

3.9.5.1.1 Engine oil operating temperature. The engine oil sump temperature shall not exceed 250 °F.

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3.9.5.1.2 Engine oil consumption. The engine oil consumption shall not exceed 0.0035 pounds per brake horsepower-hour (lbs/bhp-hr) under any operating condition.

3.9.5.2 Engine oil filter. The engine oil filter shall be in accordance with 3.13.1.4.4 of SAE ARP1247.

3.9.6 Exhaust system. The exhaust system shall be constructed of stainless steel. The muffler(s) shall be constructed of aluminized steel or stainless steel. Exhaust system outlet(s) shall be directed away from personnel accessing any control panel or equipment compartment and the engine air intake, and shall not be directed toward the ground.

3.9.7 Engine fuels and fuel system.

3.9.7.1 Engine primary fuels. The following shall be primary fuels:

a. JP-5, in accordance with MIL-DTL-5624, -40 to 125 °F ambient air temperature.

b. JP-8, in accordance with MIL-DTL-83133, -40 to 125 °F ambient air temperature.

c. F-24, in accordance with NATO AFLP-3747, -40 to 125 °F ambient air temperature.

3.9.7.2 Engine alternate fuels. The following shall be alternate fuels:

a. 1-D S15, 1-D S500, and 1-D S5000 in accordance with ASTM D975, below 60 °F ambient air temperature.

b. 2-D S15, 2-D S500, and 2-D S5000 in accordance with ASTM-D975 above 32 °F ambient air temperature.

c. TS-1 in accordance with GOST 10227-86 with JP-8 additives, -40 to 125 °F ambient air temperature.

3.9.7.3 Engine emergency fuels. The following shall be emergency fuels: Jet A, Jet A1 in accordance with ASTM D1655-05, all ambient temperatures.

Note: Jet A and Jet A1 may contain corrosion inhibitor/lubricity improver and icing inhibitors which shall result in properties equal to JP-8.

3.9.7.4 Engine fuel consumption. When operating on JP-8, the engine at full rated load shall not exceed 0.8 lbs/bhp-hr brake specific fuel consumption.

3.9.7.5 Fuel system. The fuel system shall be in accordance with 3.13.1.5.1 through 3.13.1.5.11 of SAE ARP1247 except as otherwise specified herein. The fuel system shall be constructed of materials which are compatible with the fuels listed in 3.9.7.1, and, 3.9.7.2. Copper shall not be used in the fuel system. The fuel system shall be equipped with a fuel shut-off valve(s) to prevent continuous spillage when fuel lines are disconnected for service.

3.9.7.5.1 Fuel filters. Primary and secondary fuel filters and a heated fuel/water separator shall be provided. The fuel/water separator shall include a water coalescer and a drain valve that is readily accessible by an operator or a mechanic. A combination fuel filter and fuel/water separator may be provided. Fuel filter elements shall be easily replaceable by a mechanic using nothing more than common hand tools (see 7.4.1) without loss of engine prime.

3.9.7.5.2 Fuel tank. The fuel tank shall be in accordance with 3.13.1.5.5 through 3.13.1.5.9 of SAE ARP1247. The tank shall be designed so that the spraying unit can be operated on a one degree incline in any plane. The tank shall be provided with corrosion protection and baffles. A 0.25 to

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0.375 inch nominal drain valve shall be provided for emptying fuel and sediment into a container underneath the spraying unit without removal of the tank or any other major component. The fuel tank shall have a fuel fill opening of not less than three inch inside diameter and shall be designed to drain fuel spillage overboard for collection outside the spraying unit. The fuel cap shall be equipped with a retention device to prevent loss and FOD. The fuel fill opening, fuel cap, and fuel cap retention device shall be fabricated from non-sparking material.

3.9.8 Engine diagnostic and emergency shutdown systems.

3.9.8.1 Engine diagnostic system. If the engine is equipped with an electronic control module, a diagnostic system shall be provided with a means to indicate engine faults. If the spraying unit is equipped with a diagnostic or built-in-test system, the engine diagnostic system shall be integrated with it; if not, it shall be a stand-alone system.

3.9.8.2 Engine emergency shutdown system. The engine shall be equipped with an engine emergency shutdown system consisting of a fuel cutoff solenoid activated by the following conditions:

a. Low oil pressure, less than 10 psi or in accordance with the engine manufacturer's recommendations.

b. Coolant over heat or cylinder head temperature over heat condition in accordance with the engine manufacturer's recommendations.

c. Engine over speed should the engine exceed 110 percent rated speed. This is required for engines over 100 bhp and optional for engines less than 100 bhp rated horsepower.

3.9.9 Engine operator instruments. The following instruments shall be provided to the engine operator:

a. Tachometer.

b. Coolant temperature gauge.

c. Low coolant level indicator.

d. Oil pressure gauge.

e. Fuel level gauge.

f. Hour meter.

3.10 Electrical system. The spraying unit shall have a 12 or 24-volt, negative ground electrical system in accordance with 3.13.1.2 of SAE ARP1247 except as otherwise specified herein.

3.10.1 Alternator. A single or dual alternator charging system in accordance with 3.13.1.4.9 of SAE ARP1247 shall be provided. The alternator shall be capable of restoring the energy expended during an engine start in less than 15 minutes of engine idle at -40 °F.

3.10.2 Batteries and battery compartment.

3.10.2.1 Batteries. Batteries shall be of the commercial maintenance-free sealed lead acid, starved electrolyte, gas recombination, spiral wrapped, absorbent gas mat (AGM), top post type in accordance with MIL-B-18013/1.

3.10.2.2 Battery compartment. The batteries shall be enclosed in a corrosion-resistant, weatherproof box or compartment and shall be readily accessible.

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3.10.2.3 Battery cables. The battery cables shall be sized to handle the system voltage and current levels, be clearly identified with "+" and "-" or red and black markings, and shall not be spliced.

3.10.2.4 Booster receptacle. The unit shall come equipped with a booster slaver receptacle that provides the ability to jump using 12-volt battery power from one deicer to another; if that other deicer also has a similar connector. External lugs shall also be installed in the event another deicer is unavailable.

3.10.3 Winterization system. A winterization system may be provided for starting in temperatures to -40 °F. The winterization system may include heaters for engine coolant, engine oil, and the fuel tank, as well as battery warmers. The winterization system shall be designed to operate from an external AC power source utilizing the external electrical connections (see 3.10.2.4). A 25-foot cable shall be provided. A visible or audible warning device shall notify the driver if the truck remains connected to the cable when the truck engine is started. The winterization system shall incorporate high-temperature shutoff switches to prevent overheating of any fluid or component. Any harness will be shipped with the vehicle for electrical connections to operate any winterization system.

3.11 Hydraulic system. The hydraulic system shall be in accordance with 3.13.1.3 of SAE ARP1247 except as otherwise specified herein. O-ring face seal hydraulic fittings may be used in lieu of flared fittings (see 3.13.1.3.12 of SAE ARP1247). Hydraulic fluid shall be in accordance with MIL-PRF-83282. All hydraulic system components, including the hydraulic tank, shall comply with all corrosion resistance requirements specified herein. The system shall be equipped with a hydraulic oil cooler and thermostatically controlled fan to maintain the hydraulic oil at the appropriate temperature to ensure proper operation of the unit within the operation envelope specified within this document. A low-level shutdown system shall be provided to protect the hydraulic system. The hydraulic oil reservoir shall be equipped with an externally mounted level and temperature gauge. External manual controls shall be provided in the event of electrical solenoid failure.

3.11.1 Hydraulic pressure test system. Test ports with quick disconnects shall be provided, along with a hydraulic test gauge, to allow a mechanic to quickly test for correct pressure in the various hydraulic circuits.

3.11.2 Electric-driven emergency hydraulic system. An emergency hydraulic system, driven from the 12-volt DC battery, shall be provided. This system shall enable the aerial device to be maneuvered and lowered in the event of auxiliary power supply failure.

3.12 Chassis. The spraying unit shall be built on a commercial or custom truck chassis in accordance with 3.3 of SAE ARP 1971, which has been modified to satisfy the requirements of either a Class A or Class B spraying unit configuration. The chassis shall interface with the unit frame, boom assembly, and all other components in such a manner that the entire assembly exists as an integrated whole.

3.12.1 Tow hooks. Tow hooks shall be installed on the chassis frame, or on suitable members attached directly to the chassis. There shall be two tow hooks located at the front on each side of the centerline and two located at the back on each side of the centerline. The tow hooks shall be in compliance with the requirements in 3.8.2.

3.13 Truck cab. The chassis shall include a standard truck cab with seating for vehicle driver and one passenger. The driver's position shall be on the left side of the chassis to provide the maximum

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possible view of the enclosed cab under all operating conditions. Power and heated mirrors shall be included.

3.13.1 Truck cab control panel. In addition to the standard instrument panel provided by the chassis manufacturer, a separate control panel for deicing operations shall be provided that includes the following:

a. Auxiliary power supply controls (see 3.18.2).
b. Fluid pump on/off switch(s).
c. Heater on/off switch and activation indicator light.
d. Flashing (red or amber) beacon on/ off switch.
e. Emergency stop switch (see 3.2.4) and activation indicator light.
f. Overhead window washer/wiper activation switch.
g. Gauge lights for nighttime operation.
h. Deicing & anti-icing fluid level gauges.
i. Deicing fluid level pressure gauge.
j. Deicing fluid temperature gauge.

3.13.2 Truck cab observation window. The cab roof shall be fitted with a large, sloped observation window made out of safety glass or similar material. An electric wiper for overhead viewing from either the driver or passenger's seated position shall be included.

3.13.3 Truck cab boom controls. Similar to the enclosed cab boom controls (see 3.17.2), identical boom actuation and cab rotation fully proportional joystick controls shall be installed between the driver and passenger, so that either may conveniently operate them. In accordance with 3.4.8 of SAE ARP 1971, these controls shall override the enclosed cab controls. These controls shall not operate when the boom is in the upper operational envelope, between 42 feet and 75 feet of elevation, of the Class B spraying unit.

3.14 Unit frame. All components of the deicing unit, including fluid tanks, auxiliary engine, heater, fluid pumps, and boom pedestal shall be mounted on a frame that fastens to the truck chassis. This framework shall be designed to allow for mounting on any chassis that conforms to the manufacturer's requirements and containing all the relevant modifications for either a Class A or Class B configuration.

3.15 Boom assembly. The boom assembly or aerial device shall be in accordance with ANSI/SAIA A92.2. It shall be specifically designed to achieve the maximum in maneuverability, stability, and safety for de/anti-icing operations within the specified range of either the Class A or Class B spraying unit. Under all operating conditions, the boom shall operate smoothly. The boom shall rotate 360 degrees, non-continuously in accordance with 3.4.3 of SAE ARP 1971.

3.15.1 Boom stability – Class A. The boom assembly shall be designed to support a working load per person of at least 350 pounds in the enclosed cab during mobile operations or while in any extended position. With a 700-pound load placed in the enclosed cab with the boom assembly in any extended position, with no fluid in the de/anti-icing fluid tanks, the deicing unit on level ground shall be capable of remaining stable and not overturn.

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3.15.2 Boom stability – Class B. The boom assembly shall be designed to support a working load per person of at least 250 pounds in the enclosed cab during mobile operations or while in any extended position. With a 500-pound load placed in the enclosed cab with the boom assembly in any extended position, with no fluid in the de/anti-icing fluid tanks, the deicing unit on level ground shall be capable of remaining stable and not overturn.

3.15.3 Range of platform movement – Class A. The boom assembly of the Class A spraying unit shall raise the spray nozzle to a minimum vertical height of 42 feet as measured from level ground to the floor of the enclosed cab. The boom assembly shall also have a horizontal, side reach of not less than 40 feet as measured from the fluid spray nozzle to the centerline of the turntable.

3.15.4 Range of platform movement – Class B. The boom assembly of the Class B spraying unit shall raise the spray nozzle to a minimum vertical height of 75 feet as measured from the level ground to the floor of the enclosed cab. The boom assembly shall also have a horizontal, side reach of not less than 40 feet as measured from the fluid spray nozzle to the centerline of the turntable.

3.15.5 Operational envelope/limit indicators – Class B. Situated near or within view of any set of boom controls, the Class B spraying unit shall be equipped with a light panel that indicates, via proximity sensors, whether the enclosed cab is in the lower (below 42 feet) operational envelope or the upper (42 feet or above) operational envelope. Each light panel, along with audible warning, shall also provide indication when the boom has reached or is in proximity to reach any of its operational limits, such as maximum/minimum angle, maximum extension, and the full extent of the lower envelope, signifying that stabilizer deployment is required for further extension. This system shall only allow the boom to operate in stable configurations. All audible warnings shall be clearly heard in both the enclosed and truck cabs under all operating conditions.

3.15.6 Boom controls. In addition to the enclosed cab (see 3.17.2) and truck cab (see 3.13.3) controls, hydraulic boom controls shall be provided at a ground level control station. The ground level boom controls shall be fully proportional and shall extend/retract and rotate horizontally/vertically the boom. These ground level controls shall override either the enclosed cab or truck cab boom controls.

3.16 Stabilizers. Class B deicing unit shall be equipped with outriggers or stabilizers, in accordance with 3.10.9 of SAE ARP 1247, which shall be engaged whenever the enclosed cab is in the upper operational envelope, from 42 to 75 feet in elevation. Electric over hydraulic controls for the outriggers or stabilizers shall include the following interlocked features to assure safe operation:

a. Before the stabilizers can be deployed, the truck transmission shall be in neutral and the parking brake set.

b. Before the boom can be extended into the upper operational envelope (above 42 feet), the stabilizers shall be fully deployed.

c. Before the stabilizers can be retracted, the boom shall be in the lower envelope.

d. Before the truck transmission can be placed in gear, the stabilizers shall be completely stowed. Means to override this interlock shall be provided, for emergency situations, when the truck needs to be driven and completely stowing the stabilizers is not possible.

e. Cross-port check valves shall be installed to prevent hydraulic drift of each stabilizer.

3.17 Enclosed cab. An enclosed cabin along the middle to upper portion of the boom assembly shall be provided, in full accordance with SAE ARP 5058, “Enclosed Operator’s Cabin for Aircraft

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Ground Deicing Equipment." The enclosed cab shall have sufficient room for one operator. The enclosed cab shall come equipped with externally mounted side mirrors to increase operator side and rear visibility during cab maneuvering. Means to automatically level the enclosed cab, while the boom is anywhere within its operational range, shall be included. The enclosed cab shall be capable of horizontal rotation of a minimum of 90 degrees in either direction from straightforward in the plane of the boom, providing a total range of movement of 180 degrees. All parts of the enclosed cab that may come in contact with the aircraft under normal operation shall be padded with a resilient, replaceable bumper.

3.17.1 Communication system. A means for a truck cab passenger, the truck driver, and enclosed cab operator to communicate simultaneously shall be provided. The system shall be powered with the on/off/start switch of the truck chassis. If a two-way headset communication system is utilized, it shall be in accordance with 3.3.9 of SAE ARP 5058 and 3.4.17 of SAE ARP 1971.

3.17.2 Enclosed cab controls. Easily operated, readily accessible, and consolidated controls for boom actuation, cab rotation, de/anti-icing fluid operations, and forced air operations shall be provided inside the enclosed cab. Controls shall be fully proportional, electric over hydraulic, and in accordance with 3.3.5 of SAE ARP 5058 and 3.10.6 of SAE ARP 1247, of the "dead-man" type. Variable deicing fluid flow shall be controllable, while non-variable anti-icing flow, to protect the fluid from degradation, is recommended. The cab shall include controls for wipers, washers, emergency stop, and emergency hydraulics, conveniently located within reach of the operator.

3.17.3 Anti-contact. The spraying unit shall have an operator aid system consisting of two (2) side mounted physical antenna sensors, which deploy when the boom is raised from the boom rest, and a nozzle mounted antenna sensor. If any one of the three (3) antenna sensors contact an object causing the antennas to deflect, then all boom and cab movement functions shall immediately stop. The operator must use the manual override system to move away from the object that is contacting the antenna system. A comparable anti-contact system is acceptable in lieu of the antenna system, provided the system immediately stops all boom and cab movement functions prior to contact with an object.

3.18 Auxiliary power supply. The spraying unit shall use an auxiliary power supply as a means to efficiently drive all de/anti-icing, cleaning, and decontamination components of the spraying unit separate from the main engine.

3.18.1 Auxiliary engine. If a diesel auxiliary engine is utilized, it shall be in accordance with SAE ARP 1052, *Selection Criteria for Internal Combustion Engines Used in Ground Support Equipment*. The engine capacity shall be such that the continuous duty horsepower input required by all engine-driven items shall not exceed 75 percent of the full throttle rating of the engine, at sea level and 59 °F. The engine shall have a minimum of four cylinders.

3.18.1.1 Auxiliary engine exhaust. The auxiliary engine exhaust shall be routed and properly guarded so as to eliminate any possibility of danger to personnel, components, and wiring. The exhaust shall terminate at the rear of the spraying unit no less than 95 inches above the ground, and shall be equipped with a spark-arresting muffler.

3.18.2 Auxiliary power supply controls. All auxiliary power supply controls shall be mounted on the truck cab control panel. Auxiliary power supply controls, indicators, and gauges shall include: off/on/start switch, fuel level gauge, tachometer, oil pressure gauge, water temperature gauge, battery charge indicator gauge, and hour meter.

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3.18.3 Auxiliary power supply fuel tank. In addition to the standard truck chassis fuel tank, a separate fuel tank with a minimum capacity of 150 gallons shall be provided. The fuel system shall be capable of operating the auxiliary power supply and heater at rated output for a minimum of four hours without refueling. The fuel system shall be designed to provide for shutdown of the heater first, auxiliary power supply second, and chassis engine last. The chassis fuel tank and the 150 gallon fuel tank shall be connected together to provide additional capacity.

3.19 Plumbing. The de/anti-icing system shall be designed to be compatible with all types of deicing and anti-icing fluids, as well as, cleaning compound solutions conforming to MIL-PRF-87937. Plumbing hard lines and fittings shall be non-corrosive and wherever possible stainless steel.

3.20 Pumping systems. Two independently plumbed pumping systems shall be provided to drive heated deicing fluid and unheated anti-icing fluid to the delivery system.

3.20.1 Deicing fluid pump. A pump shall be provided for the delivery of deicing fluid from the deicing fluid tank, through the heater, and to either the spray nozzle of the enclosed cab or the ground hand nozzle on the front bumper of the chassis. The pump shall be capable of delivering heated deicing fluid from the enclosed-cab nozzle at a flow rate and pressure sufficient enough to deice the aircraft but not damage the surface.

3.20.2 Anti-icing fluid pump. A pump which will minimize degradation to the anti-icing fluid shall be provided. The pump shall be capable of delivering unheated anti-icing fluid to the spray nozzle of the enclosed cab at a flow rate and pressure sufficient enough to apply fluid but not high enough to degrade (being broken down or diffused enough to be rendered ineffective) the anti-icing fluid. Pump shall shut off when spray nozzle is closed to further ensure limited degradation of the anti-icing fluid.

3.20.3 Fluid refill system. The anti-icing fluid pump system shall include the means to reverse flow, so that the anti-icing fluid tank may be filled from 55-gallon drums with the anti-icing fluid pump providing the necessary suction. The same such system shall be made available when specified for refilling deicing fluid tanks. The systems shall refill the fluid tanks at a minimum rate of 18 gallons per minute and without degradation to the fluids being refilled.

3.21 Fluid heating system. A heater that operates on either diesel fuel, Jet A or JP-8, or other power source shall be provided to heat deicing fluid. It shall have fully automatic ignition and controls for smokeless operation under all conditions, including starting and shut down. The fluid heater system shall have a minimum output heat capacity of 2,000,000 Btu per hour and shall provide maximum nozzle temperature in less than five (5) minutes when the fluid temperature starts at 50 °F. It shall be equipped with a thermostat that maintains a fluid temperature of 185 ±10 °F, and an automatic shutdown that shall activate when the fluid reaches an excessive temperature.

3.21.1 Heater ignition. The heater ignition shall operate at temperatures as low as -40 °F. It shall be in accordance with NFPA 70 and ignition cables within the system shall be in accordance with UL 558.

3.21.2 Combustion air blower. If applicable, a combustion air blower, powered by the auxiliary power supply pump, shall be provided as an integral part of the heater system. This blower, which shall automatically adjust in speed in response to the heater output requirements, shall provide the proper amount of air for fuel/air mixture.

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3.21.3 Heater exhaust. The heater exhaust shall be directed up and to the rear of the unit. The exhaust outlet shall be protected from precipitation. It shall terminate no less than 95 inches vertically from the ground. The gas duct shall not be constructed of flexible metal tubing.

3.21.4 Heater control. Electrical controls shall be centrally located at a heater control panel, with the exception of the thermal switches that may be located on the heating unit and the truck cab controls (see 3.13.1). The heater control panel shall include a method of displaying fluid inlet and outlet temperature, fuel pressure, combustion air pressure, and fluid flow. These parameters shall be capable of being monitored when the heater is idling or in operation.

3.21.5 Heater safety. The following safety features shall be incorporated into the heating system:

a. The ignition system shall prevent discharge of fuel until proper ignition is assured.

b. The heater during operation shall not be adversely affected by vibration, sudden shock, or tilted operation.

c. The heater shall include a flow control switch to shut off fuel flow to the burner when there is no fluid flow through the heater.

d. A safety valve shall automatically relieve pressure in the heater coil when the normal operating pressure is precariously exceeded.

e. The heater shall shut down when exhaust temperatures are excessive.

3.22 Fluid tanks. De/anti-icing fluid tanks shall be durably constructed of stainless steel. While the tanks may be of integrated construction, the deicing tank and anti-icing tank shall be physically and thermally isolated. The tanks shall be vented to allow for expansion and shall have internal perforated baffles to prevent fluid impact or sloshing during movement or rapid stopping. The tanks shall have sufficient-size openings to allow access to a mechanic performing repairs or maintenance, and an easily accessible fluid sampling port to collect samples of the deice/anti-ice fluids in the tank. The tanks shall be equipped with a sump and holes for intake and return lines. The deicing fluid tank shall also be equipped with a strainer.

3.22.1 Fluid tank capacity – Class A. For Class A spraying units, the total fluid capacity shall be a minimum of 1,600 gallons, in accordance with 3.2.5 of SAE ARP 1971. The deicing fluid tank shall have a minimum fluid capacity of 1,450 gallons, and the anti-icing fluid tank shall have a minimum fluid capacity of 150 gallons.

3.22.2 Fluid tank capacity – Class B. For Class B spraying units, the total fluid capacity shall be a minimum of 2,600 gallons. The deicing fluid tank shall have a minimum capacity of 2,250 gallons, while the anti-icing fluid tank shall have a minimum capacity of 350 gallons.

3.22.3 Tank-level indicators. A gauge indicating the level of the deicing fluid shall be provided inside of the truck cab, along with a “Low Level” indicator that registers when the deicing fluid is too low to continue operation and must shut down to prevent damage to the deicing pump. Indicator lights activated by an in-tank sensor shall provide the operator with a clear visual signal indicating the de/anti-ice, and/or water tank is full. Additionally, sight gauges shall be provided on each de/anti-ice tank to show the fluid level.

3.22.4 Tank drain handles. Drain handles shall be located to the side of the spraying unit to allow for easy opening and closing of tank drain valves.

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3.23 Delivery system. Operation of the delivery system, consisting of the de/anti-icing fluid spray nozzle, forced air system, and ground handline system shall be independent and not be contingent upon operation of the heater or the boom. All components of the delivery system shall withstand the working pressures specified and a hydrostatic pressure of at least twice the specified working pressure without leakage or failure.

3.23.1 De/anti-icing fluid spray nozzle. A dual fluid spray nozzle shall be capable of variably delivering deicing fluid up to a minimum of 40 gallons per minute with the boom fully elevated. The nozzle shall also be capable of delivering anti-icing fluid up to a minimum of 20 gallons per minute without fluid degradation with the boom fully elevated. Various spray patterns, to include a wide spread fine mist, a heavy conical spray, and a solid straight stream, shall be able to be selected from the controls.

3.23.2 Forced air system. A forced air system shall be utilized that it is capable of continuously supplying air to the forced air nozzle. The air outlet shall be such that it will prevent the ingestion or discharge of internal components or damaging fragments in the event of a catastrophic failure of the system.

3.23.2.1 Forced air nozzle. A separate nozzle to provide forced air, piped from the air compressor shall be provided. The air shall be delivered from the nozzle at a volumetric flow rate of no less than 1,340 cubic feet per minute and at a velocity of 700 miles per hour. The pressure of the air emitted at the nozzle shall not be in excess of 13 pounds per square inch.

3.23.2.2 Air system meter. An easily accessible hour-meter shall be provided to monitor the hours the forced air nozzle is in operation.

3.23.2.3 Reduced flow. The system shall include a means to apply deicing fluid in a reduced flow setting for lighter ice loads, such as frost, at a maximum flow rate of 10 gallons per minute.

3.23.3 Control and range of nozzle motion. A control system within the enclosed cab shall remotely control both the de/anti-icing fluid and forced air nozzles, which may be joined together and move in unison. Both nozzles shall have a minimum horizontal range of movement of 90 degrees to the right and left of the straight-ahead position, providing a total range of movement of 180 degrees. The vertical range of movement shall be 90 degrees up and a minimum of 45 degrees down from the horizontal position. The nozzles shall operate at a minimum speed of 5 degrees of arc per second.

3.23.4 Ground hand-line system. A hose reel shall be mounted to the unit for storing a ground delivery hose, with a combo-nozzle for quick disconnects. The hose shall be 50-foot in length on the Class A unit, and 100-foot in length on the Class B unit. A shut-off and drain valve shall be provided to permit draining and removal of the hose reel without interfering with the operation of the rest of the system. The reel shall be capable of retracting the hose unassisted by operators through mechanical or electrical force. If implemented, the rewind switch shall be adjacent to the hose reel and protected from interference with the hose. The reel shall be equipped with a manual rewind hand crank, and brake.

3.23.5 Fluid usage meter. The spraying unit shall be equipped with a fluid usage meter for the deicing fluid and anti-icing fluid separately, which can be used to store volumetric history. A display shall be installed in full view of the enclosed cab operator. This display shall be clearly visible under all operating conditions.

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3.24 Rear camera. The spraying unit shall be equipped with a dual camera video system. One camera shall be a rear-facing camera mounted on the rear of the spraying unit, and the second camera shall be mounted on the rear of the enclosed cab. The monitor shall be mounted in the enclosed cab and shall automatically switch to both cameras when the operator moves the joystick to the rear.

3.25 Single operator drive system. When specified (see 7.2.e), the spraying unit shall be able to drive as well as control all critical deicer functions from the enclosed cab; with only a single operator. The deicer can be operated in either single, or dual operator modes in this configuration.

3.26 Lighting. All lighting shall be of LED type and have easily procurable replacement bulbs and fixtures.

3.26.1 Floodlights. Exterior non-glare white lighting shall be provided for night operation. Lights shall be positioned to provide adequate working light at ladder(s), access step(s), and work area(s). Lights can be manually activated by operator.

3.27 Workmanship. The spraying unit including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include the following: freedom from blemishes, defects, burrs, sharp corners and edges, accuracy of dimensions, surface finish, radii of fillets, thoroughness of welding, painting, riveting, marking of parts and assemblies, proper alignment of parts, and tightness of assembly fasteners.

3.27.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.27.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.27.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

3.27.4 Cleaning. The spraying unit shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

3.28 Product familiarization video. When specified (see 7.2.f), a product familiarization video shall be provided to aid visually the acclimation of a user to the features present on the spraying unit.

4. REGULATORY REQUIREMENTS.

4.1 Recycled recovered materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA

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designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS.

The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The government reserves the right to require proof of such conformance.

5.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production inspection (see 5.2).

b. Conformance inspection (see 5.3).

5.2 First production inspection. The first production spraying unit shall be subjected to the analyses, demonstrations, examinations, and tests described in 5.5.1 through 5.5.12. The contractor shall provide or arrange for all test equipment and facilities. The contractor is to receive DoD test rates at a Major Range Test Facility Base. Except as otherwise specified, all testing in which the engine is operated shall be performed using JP-8 or F-24 turbine fuel.

5.3 Conformance inspection. Each production spraying unit shall be subjected to the examination described in 5.5.1

5.4 Inspection requirements.

5.4.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

5.4.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity.

5.4.3 Test rejection criteria. Throughout all tests specified herein, the spraying unit shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

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c. Degradation of anti-icing fluid.

d. Structural failure of any component, including permanent deformation, or evidence of impending failure.

e. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturers (OEM's) specifications or tolerances shall be utilized for making a determination.

f. Evidence of corrosion or deterioration.

g. Misalignment of components.

h. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

i. Interference between the spraying unit components or between the spraying unit, the ground, and all required obstacles, with the exception of normal contact by the tires.

j. Evidence of undesirable mobility characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

k. Overheating of the engine, transmission, or any other vehicle component.

l. Shutdown faults from:

    i. Engine cooling system.

    ii. Engine lubrication system.

    iii. Engine protective circuits.

5.5 Test methods.

5.5.1 Examination of product. Each spraying unit shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 5.5. A contractor-generated, government-approved checklist shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each spraying unit function shall be verified. Certifications and analyses shall be provided in accordance with Table III. Each production spraying unit shall be inspected to a government-approved reduced version of the checklist.

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TABLE III. Certifications and analyses.

Paragraph	Required Certifications and Analyses
3.3.3.2 Snow	Contractor analysis of the snow load requirement (see 5.5.3.3.2).
3.5.2 Air transportability	Contractor air transportability analysis (see 5.5.5.1). Air Transportability certification shall be coordinated with the government and Air Transportability Test Loading Activity.
3.5.3 Tie downs	Contractor tie down provision analysis (see 5.5.5.3).
3.9 Engine and related equipment	Engine manufacturer certification that the engine is in accordance with all applicable requirements, including exhaust emissions standards and fuels. Engine manufacturer application approval for the engine and its installation, including cooling system, lubrication system, and mounting system.

5.5.2 Sound level test. A first production spraying unit shall be tested for compliance to the recommended practice in SAE J 1096 indicated in 3.2.4.3.

5.5.3 Environmental testing.

5.5.3.1 High temperature storage and operation test. A first production spraying unit shall be tested in accordance with MIL-STD-810, Method 501.5, Procedures I and II, to demonstrate compliance with the high temperature storage and operating requirements of 3.3.1 and 3.3.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

5.5.3.2 Low temperature storage and operation test. A first production spraying unit shall be tested in accordance with MIL-STD-810, Method 502.5, Procedures I and II, to demonstrate compliance with the low temperature storage and operating requirements of 3.3.1 and 3.3.2, as well as the engine starting requirement of 3.9.2 and the alternator charging requirement of 3.10.1. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

5.5.3.3 Precipitation.

5.5.3.3.1 Rain test. A first production spraying unit shall be tested in accordance with MIL-STD-810, Method 506.5, Procedure I, to demonstrate compliance with 3.3.3.1.

5.5.3.3.2 Snow load analysis. An engineering analysis shall be performed to demonstrate compliance with the snow load requirement of 3.3.3.2, using a specific gravity of snow of 0.1 (Ref. 5.3 of MIL-STD-810).

5.5.3.3.3 Ice accretion test. A first production spraying unit shall be tested in accordance with MIL-STD-810, Method 521.3 with an ice thickness of 1.5-inches to demonstrate compliance with

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the ice accretion requirement of 3.3.3.3. The contractor shall identify those areas of the spraying unit where ice removal is required prior to operation.

5.5.4 Weight and dimension tests.

5.5.4.1 Weight and center of gravity test. The weight, center of gravity, and axle weights of a first production spraying unit shall be measured to demonstrate compliance with the weight requirement of 3.4 and the axle weight requirement of 3.5.2.2.

5.5.4.2 Dimension measurement. A first production spraying unit shall be measured to demonstrate compliance with the dimensional requirements of 3.4.

5.5.5 Transportability verification.

5.5.5.1 Air transportability analysis. An engineering analysis shall be performed to demonstrate compliance with the air transportability requirements of 3.5.2. The analysis shall include the tie downs and all major components and their ability to withstand the accelerations specified in 3.5.2. The evaluation shall also include a dimensional analysis for the spraying unit while traversing the ramp and while loaded aboard C-5 and C-17 aircraft.

5.5.5.2 Equipment removal and reconfiguration demonstration. A first production spraying unit shall be configured for transport on C-5 and C-17 aircraft and then reconfigured for operation to demonstrate compliance with 3.5.2.4. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

5.5.5.3 Tie down provision verification.

5.5.5.3.1 Tie down provision analysis. An engineering analysis shall be performed to demonstrate compliance with the tie down provision requirements of 3.5.3.

5.5.5.3.2 Tie down provision test. A first production spraying unit shall be tested to demonstrate compliance with the tie down provision requirements of 3.5.3.

5.5.6 Maintainability demonstration. A first production spraying unit shall demonstrate compliance with 3.6. All recommended preventive maintenance tasks shall be performed and the task times shall be recorded. The recommended frequencies of the preventive maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of preventive maintenance time per measure of use, such as calendar time or hours of operation. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472. All preventive maintenance tasks recommended to be performed daily and at the routine PMI shall also be performed by personnel wearing arctic mittens.

5.5.7 Mobility tests. A first production spraying unit shall be tested in accordance 4.6.9 of SAE ARP1247 to demonstrate compliance with 3.8.

5.5.8 Brake tests.

5.5.8.1 Parking brake test. A first production spraying unit shall be tested to demonstrate compliance with 3.13.1.7.2.3 of SAE ARP1247 and 3.8.7.1.

5.5.8.2 Service brake test. A first production spraying unit shall be tested to demonstrate compliance with 3.13.1.7.2 of SAE ARP1247 and 3.8.7.2.

5.5.9 Boom stability test. A first production spraying unit shall be operated to demonstrate compliance with 3.15. With 700 pounds placed in the enclosed cab of a Class A spraying unit and

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500 pounds placed in the enclosed cab of a Class B spraying unit on level ground, the boom shall be extended to the maximum position with no fluid in the de/anti-icing fluid tanks; the unit shall remain stable and shall not overturn. Boom certifications shall be provided to the government and paper copies shall accompany the vehicle prior to shipment. A certification placard shall be affixed to the vehicle.

5.5.10 Pump and fluid delivery test. A first production spraying unit shall be tested to demonstrate conformance of unit characteristics with all fluid flow values within 3.20.3, 3.23.1, and 3.23.2.3.

5.5.11 Welds and welding. Welds shall be inspected to ensure proper penetration and verified to be in strict accordance with the Visual Inspection criteria set forth in the AWS Codes. The inspection shall be performed by an AWS Certified Weld Inspector (CWI) during and after the construction of all spraying units. Documentation of all inspections for the first production spraying unit shall be provided to the government. The government reserves the right to require verification inspection at any time during construction in accordance with 6.1.2.2 of AWS D1.1, 5.1.1 of AWS D1.2, and Non-specified Non-Destructive Test (NDT) other than visual in accordance with 6.6.5 of AWS D1.1. All welding documentation for subsequent production units shall be made available to the government at any time during the life of the contract pursuant to AWS D1.1 and D1.2.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 Department of Defense and Federal documents. Department of Defense and Federal documents, except for GOST 10227-86, are available online at https://quicksearch.dla.mil/. A copy of GOST 10227-86 can be obtained from the Procuring Contracting Officer (PCO).

7.1.2 Federal Acquisition Regulations (FAR). Copies of documents may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained at https://www.ecfr.gov/.

7.1.3 American Society for Testing and Materials (ASTM). Application for copies should be addressed to ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959 or online at http://www.astm.org/.

7.1.4 American Welding Society (AWS). Copies of standards may be obtained at https://pubs.aws.org /.

7.1.5 Occupational Safety and Health Association (OSHA). Application for copies should be addressed to 200 Constitution Ave., Washington D.C. 20210 or online at http://www.osha.gov/.

7.1.6 Society of Automotive Engineers (SAE). Copies of standards may be obtained at http://www.sae.org/ or from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.

7.1.7 Scaffold & Access Industry Association (ANSI/SAIA). Copies of standards may be obtained at http://www.saiaonline.org/ or from Scaffold & Access Industry Association, 1617 East Rairvan Lane, Suite 5, P.O. Box 20574, Phoenix, AZ 05034.

7.1.8 Underwriters Laboratories (UL). Copies of documents may be obtained at http://www.ul.com/ or from the Underwriters Laboratories, 333Pfingsten Road, Northbrook, IL 60062-2096.

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7.1.9 National Fire Protection Association (NFPA). Copies of documents may be obtained at http://www.nfpa.org/ or from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02169-7471.

7.1.10 The Code of Federal Regulation (CFR). Copies of regulations may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402 or online at http://www.gpo.gov/fdsys/search/home.action.

7.1.11 (ITSDF) Industrial Truck Standards Development Foundation. Copies of standards may be obtained from www.itsdf.org or the Industrial Truck Standards Development Foundation Suite 460, 1750 K Street NW, Washington, DC 20006.

7.1.12 (TRA) The Tire and Rim Association, Inc. Copies of standards may be obtained from http://www.us-tra.org/ or from the Tire and Rim Association, 175 Montrose West Ave., Suite 150, Copley, OH 44321.

7.1.13 (ATTLA) Air Transportability and Test Loading Agency. ATTLA may be contacted at ATTLA@us.af.mil.

7.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.

b. If overseas transports provisions are required (see 3.2.1.1.1).

c. Finish color required if not Semi-gloss Green (see 3.2.1.3).

d. If any special test equipment or software shall accompany the vehicle (see 3.6.5).

e. If a single operator drive system is required by the user (see 3.25).

f. If a product familiarization video is required (see 3.28).

g. Packaging requirements (see 6).

7.3 National Stock Number (NSN). The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with this CID.

NSN	SPRAYING UNIT
1730-00-555-6205	CLASS A
1730-01-495-5449	CLASS B

7.4 Definitions.

7.4.1 Common hand tool. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open end, boxed end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex key wrenches, in SAE sizes up to and including 1-inch and metric sizes up to and including 25-mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

7.4.2 Special tool. A tool that is not commercially and readily available from a source other than the spraying unit contractor.

7.5 Key words.

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Boom and enclosed cabs.

Decontaminating.

Extended reach.

Heated and unheated.

Deicer.

7.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this CID based on the entire content irrespective of the marginal notations.

Concluding Material:

Custodians:
Air Force 184

Preparing Activity:
Air Force 184

Reviewer:
AF-84

Agent:
AF-110

(Project 1730-2020-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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